

E-Commerce Dashboard with On-Page SEO using React Helmet

1. Introduction

NovaCart is an online grocery store marketplace that allows customers to purchase fresh groceries online.

The platform provides a convenient shopping experience with product browsing, secure payments, order tracking,

and fast delivery. This assignment focuses on designing an Admin Dashboard for managing the store and implementing

Search Engine Optimisation (SEO) strategies using database fields and React Helmet.

2. Project Overview

The NovaCart platform is built using the MERN Stack:

Frontend: React.js

Backend: Node.js with Express.js

Database: MongoDB

Authentication: JWT

Payment Integration: Stripe or PayPal

The system contains an admin dashboard that allows administrators to manage products, orders, users, and SEO settings.

SEO is important because it improves search engine ranking and helps customers discover the store easily.

3. Admin Dashboard Design

The Admin Dashboard allows administrators to manage the complete e-commerce system.

Main Modules:

- Product Management
- Order Management
- Inventory Management
- Customer Management
- SEO Management
- Analytics Dashboard

Example Dashboard Layout:

Sidebar Navigation

- Dashboard
- Products
- Orders
- Customers
- SEO Settings
- Reports

Main Panel

- Sales Overview
- Orders Summary
- Inventory Alerts
- Website Traffic

Dashboard

Products

Orders

Customers

Back to Store

Dashboard Overview

Total Revenue

\$24,562

Total Orders

1,245

Total Customers

892

Products

1,452

Recent Revenue

[Chart Visualization Placeholder]

Recent Orders

Order #1021

2 mins ago

\$45.00

Order #1022

2 mins ago

\$45.00

Order #1023

2 mins ago

\$45.00

Order #1024

2 mins ago

\$45.00

Dashboard

Products

Orders

Customers

Products

Add New Product

Basic Information

Product Name

Apple

Category

Fruits

Price (\$)

23

Stock Quantity

9

SEO Optimization

URL Slug

e.g., fresh-organic-tomatoes

The URL friendly identifier for the product page.

Meta Title

Title shown in search engines

Search for fresh groceries...

Log In

Welcome Back

Email Address

Password

Log In

Don't have an account? Sign Up

Your one-stop destination for fresh groceries, daily

Quick Links

About Us

Customer Services

My Account

Subscribe

Get the latest updates on new products and upcoming sales.

4. Database Design for SEO (On-Page SEO Fields)

To improve search engine visibility, specific SEO fields must be stored in the database.

Example MongoDB Product Schema:

title

slug

description

price

category

image

SEO Fields:

metaTitle

metaDescription

metaKeywords

canonicalURL

ogTitle

ogDescription

ogImage

These fields help search engines understand the page content and display better results in search listings.

5. Example MongoDB Schema

```
const productSchema = new mongoose.Schema({
```

```
  title: String,
```

```
  slug: String,
```

```
description: String,  
price: Number,  
category: String,  
image: String,  
  
metaTitle: String,  
metaDescription: String,  
metaKeywords: [String],  
canonicalURL: String,  
ogTitle: String,  
ogDescription: String,  
ogImage: String  
});
```

6. Implementing On-Page SEO using React Helmet

React Helmet allows developers to manage meta tags inside React components.

These tags are important for search engines and social media previews.

Example React Component:

Here are all the **React Helmet** code snippets from your NovaCart project:

1. Setup

```
import { HelmetProvider } from 'react-helmet-async'  
  
createRoot(document.getElementById('root')).render(  
  <StrictMode>  
    <HelmetProvider>      {/* ← Wraps whole app */}  
      <BrowserRouter>
```

```
    <App />

  </BrowserRouter>

  </HelmetProvider>

  </StrictMode>

)
```

2. Home Page

```
import { Helmet } from 'react-helmet-async';
```

```
const Home = () => {

  return (

    <main>

      <Helmet>

        <title>NovaCart - Fresh Groceries Delivered Quickly</title>

        <meta name="description" content="Shop fresh, organic groceries online at NovaCart.
Lightning-fast delivery straight to your doorstep." />

        <meta name="keywords" content="online grocery, fresh fruits, organic vegetables, grocery
delivery, NovaCart" />

      </Helmet>

      {/* rest of page */}

    </main>

  );

};
```

3. Shop Page

```
<Helmet>

  <title>Shop Fresh Organic Groceries | NovaCart</title>
```

<meta name="description" content="Browse our complete catalog of farm-fresh fruits, crisp vegetables, dairy, meats, and household essentials." />

<meta name="keywords" content="buy groceries online, fresh food catalog, shop organic products, NovaCart store" />

</Helmet>

4. About Page

<Helmet>

<title>About Us | NovaCart</title>

<meta name="description" content="Learn about NovaCart's mission to bring fresh, organic, and locally-sourced groceries straight to your doorstep." />

<meta name="keywords" content="about NovaCart, our mission, fresh groceries, local farmers, organic food delivery" />

</Helmet>

5. Contact Page

<Helmet>

<title>Contact Us | NovaCart</title>

<meta name="description" content="Get in touch with the NovaCart support team. We're here to help with your orders and deliveries." />

<meta name="keywords" content="contact NovaCart, customer support, help center, grocery delivery help" />

</Helmet>

6. Categories Page

<Helmet>

<title>Browse Categories | NovaCart</title>

<meta name="description" content="Explore a wide variety of grocery categories at NovaCart including fresh fruits, vegetables, dairy, bakery items, and more." />

```
<meta name="keywords" content="grocery categories, fresh fruits, vegetables, dairy products, bakery, household goods" />
```

```
</Helmet>
```

7. Admin Products Page

```
<Helmet>
```

```
<title>Products Management | NovaAdmin</title>
```

```
</Helmet>
```

```
...
```

Each page gets its own unique title and description, which is the core of On-Page SEO!

7. React Helmet Code

```
import { Helmet } from "react-helmet";
```

```
function ProductPage({ product }) {
```

```
  return (
```

```
    <>
```

```
    <Helmet>
```

```
      <title>{product.metaTitle}</title>
```

```
      <meta name="description" content={product.metaDescription} />
```

```
      <meta name="keywords" content={product.metaKeywords.join(",")} />
```

```
      <link rel="canonical" href={product.canonicalURL} />
```

```
      <meta property="og:title" content={product.ogTitle} />
```

```
      <meta property="og:description" content={product.ogDescription} />
```



```
<meta property="og:image" content={product.ogImage} />
```

```
</Helmet>
```

```
<h1>{product.title}</h1>
```

```
<p>{product.description}</p>
```

```
</>
```

```
);
```

```
}
```

8. SEO Benefits

Using SEO fields and Helmet provides the following advantages:

- Improves Google ranking
- Better search visibility
- Rich previews on social media
- Higher click-through rate
- Better user experience